# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

# CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Objective Type (Group-I)

**Time Allowed:** 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The wave number of the light emitted by a certain source is 2 × 10^6 m^-1 the wavelength of this light will be:

(A) 500 nm
(B) 500 m
(C) 200 nm
(D) 5 × 10^7 m

Q2. Which of the following will have the same number of molecules at S.T.P:

(A) 280 cm^3 of CO2 and 280 cm^3 of N2O
(B) 11.2 dm^3 of O2 and 32 g of O2
(C) 44 g of CO2 and 11.2 dm^3 of CO
(D) 28 g of N2 and 5.6 dm^3 of oxygen

Q3. The change in heat energy of a chemical reaction at constant temperature and pressure is called:

(A) Enthalpy change
(B) Bond energy
(C) Heat of sublimation
(D) Internal energy change

Q4. One mole of SO2 contains:

(A) 6.02 × 10^23 atoms of oxygen
(B) 18.1 × 10^23 molecules of SO2
(C) 6.02 × 10^23 atoms of sulphur
(D) 4 gram atoms of SO2

Q5. With increase of 10°C temperature, the rate of reaction doubles, this increase in rate of reaction is due to:

(A) Decrease in activation energy of reaction
(B) Decrease in the number of collisions between reactant molecules
(C) Increase in activation energy of reactants
(D) Increase in number of effective collisions

Q6. NH3 shows maximum boiling point among the hydrides of group V-A elements due to:

(A) Very small size of nitrogen
(B) Lone pair of electrons present on nitrogen
(C) Enhanced electronegative character of nitrogen
(D) Pyramidal structure of NH3

Q7. Which one does not undergo sublimation:

(A) Ammonium chloride
(B) Naphthalene
(C) Iodine
(D) Mercury

Q8. The mass of one mole of electrons is:

(A) 1.008 mg
(B) 0.55 mg
(C) 0.184 mg
(D) 1.673 mg

Q9. Which one has a regular tetrahedral shape:

(A) SnCl2
(B) CH4
(C) SO3
(D) BF3

Q10. Molarity of pure water is:

(A) 1
(B) 18
(C) 55.5
(D) 6

Q11. Splitting of spectral lines when atoms are subjected to strong electrical field is called:

(A) Zeeman effect
(B) Stark effect
(C) Photoelectric effect
(D) Compton effect

Q12. The molar volume of CO2 is maximum at:

(A) S.T.P
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2 atm

Q13. Which of the following statements is not correct about galvanic cell:

(A) Anode is negatively charged
(B) Reduction occurs at anode
(C) Cathode is positively charged
(D) Reduction occurs at cathode

Q14. Which of the following is a pseudosolid:

(A) CaF2
(B) Glass
(C) NaCl
(D) All

Q15. Solvent extraction method is particularly useful technique for separation, when the product to be separated is:

(A) Non-volatile or thermally unstable
(B) Volatile or thermally stable
(C) Non-volatile or thermally stable
(D) Volatile or thermally unstable

Q16. The number of bonds in nitrogen molecule is:

(A) One σ and one π
(B) One σ and two π
(C) Three sigma only
(D) Two σ and one π

Q17. For which system does the equilibrium constant, Kc has the units of (concentration)^-1:

(A) $N_2 + 3H_2 \rightleftharpoons 2NH_3$
(B) $H_2 + I_2 \rightleftharpoons 2HI$
(C) $2NO_2 \rightleftharpoons N_2O_4$
(D) $2HF \rightleftharpoons H_2 + F_2$

## Essay Type (Group-I)

**Time Allowed:** 2.40 hrs. | **Maximum Marks:** 68

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Why we use the term relative atomic mass?

ii Many chemical reactions take place in our surroundings involve limiting reactant. Give reason.

iii How can you justify with example that molecular formula = n × empirical formula

iv In solvent extraction why repeated extraction using small portion of solvent is more efficient than single extraction using large volume of solvent.

v How the decolorization of crude crystal can take place?

vi What is the purpose of Gooch Crucible?

vii Give characteristics of plasma.

viii What are the faulty points of kinetic molecular theory of gas?

ix Water vapours do not behave ideally at 273 K. Give reason.

x Give applications of common ion effect (any two).

xi How do the buffer acts?

xii Solubility of glucose in water is increased by increasing temperature. Give reason.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i What is polarizability? Give its relation with London dispersion forces.

ii Why H2O is liquid but NH3 is gas at room temperature?

iii Why graphite conduct electricity in one direction only not in other?

iv What is habit of crystal? How it is changed?

v Why positive rays are called canal rays?

vi How neutrons were discovered?

vii Give difference between continuous spectrum and line spectrum.

viii What are slow and fast neutrons?

ix What is continuous solubility curve? Which solution give this type of curve?

x Why 1 molal solution of NaOH is dilute as compared to its 1 molar solution?

xi What is order of reaction? Give examples.

xii What do you mean by rate determining step? Give example.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Why the size of a cation is smaller as compared to its parent atom?

ii What is octet rule? Give one example.

iii Define co-ordinate covalent bond, give one example.

iv Dipole moment of CO2 is zero but SO2 is 1.61 D. Why?

v Define thermochemistry.

vi Define enthalpy of formation. Give example.

vii Describe enthalpy of neutralization by taking example of HCl and NaOH.

viii Describe Nickle Cadmium Cell.

ix Define anode and cathode.

QSECTION-II. Note: Attempt any THREE questions.

QQ.5(a). How can the percentage of carbon, hydrogen and oxygen in the given organic compound be estimated by combustion analysis?

QQ.5(b). Derive an expression to calculate the radius of revolving electron in the nth orbit of hydrogen atom.

QQ.6(a). A sample of nitrogen gas is enclosed in a vessel of volume 380 cm^3 at 120°C and pressure of 101325 Nm^-2. This gas is transferred to a 10 dm^3 flask and cooled to 27°C. Calculate the pressure in Nm^-2 exerted by a gas at 27°C.

QQ.6(b). Explain the structure and function of voltaic or galvanic cell.

QQ.7(a). Explain type of hybridization in H2O and NH3.

QQ.7(b). State first law of thermodynamics. Also prove that ΔE = qv

QQ.8(a). Write four properties of covalent solids.

QQ.8(b). The solubility of CaF2 in water at 25°C is found to be 2.05 × 10^-4 mol dm^-3. What is the value of Ksp at this temperature?

QQ.9(a). Describe a method to determine the boiling point elevation of a solution.

QQ.9(b). Define order of reaction. Describe it with three examples.

## GROUP – II

### Objective Type (Group-II)

**Time Allowed:** 20 Min. | **Maximum Marks:** 17

> Note: Inter, Part I, 2022 Lahore Board. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Isotopes differ in:

(A) Properties which depend upon mass
(B) Arrangement of electrons in orbitals
(C) Chemical properties
(D) The extent to which they may be affected by electromagnetic field

Q2. The mass of one mole of electron is:

(A) 1.008 mg
(B) 0.55 mg
(C) 0.184 mg
(D) 1.673 mg

Q3. The comparative rates at which the solutes move in paper chromatography depends on:

(A) The size of the paper
(B) Rf values of solutes
(C) Temperature of the experiment
(D) Size of the chromatographic tank used

Q4. Chromatography in which the stationary phase is a solid classified as:

(A) Partition chromatography
(B) Gas chromatography
(C) Adsorption chromatography
(D) Thin layer chromatography

Q5. Which of the following will have the same number of molecules at S.T.P.:

(A) 280 cm^3 of CO2 and 280 cm^3 of N2O
(B) 11.2 dm^3 of O2 and 32 g of O2
(C) 44 g of CO2 and 11.2 dm^3 of CO
(D) 28 g of N2 and 5.6 dm^3 of oxygen

Q6. How should the condition be changed to prevent the volume of a given gas from expanding when its mass is increased:

(A) Temperature is lowered and pressure is increased
(B) Temperature is increased and pressure is lowered
(C) Temperature and pressure both are lowered
(D) Temperature and pressure both are increased

Q7. London dispersion forces are the only forces present among the:

(A) Molecules of water in liquid state
(B) Atoms of helium in gaseous state at high temperature
(C) Molecules of solid iodine
(D) Molecules of hydrogen chloride gas

Q8. Diamond is a bad conductor because:

(A) It has a tight structure
(B) It has a high density
(C) There are no free electrons present in the crystal of diamond to conduct electricity
(D) is transparent to light

Q9. Quantum number values for 2p orbitals are:

(A) n = 2, l = 1
(B) n = 1, l = 2
(C) n = 1, l = 0
(D) n = 2, l = 0

Q10. Bohr's model of atom is contradicted by:

(A) Plank's quantum theory
(B) Dual nature of matter
(C) Heisenberg's uncertainty principle
(D) All of these

Q11. The number of bonds in nitrogen molecule is:

(A) one σ and one π
(B) one σ and two π
(C) Three sigma only
(D) Two σ and one π

Q12. The bond angle in NH3 molecule is:

(A) 109.5°
(B) 107.5°
(C) 104.5°
(D) 108°

Q13. The net heat change in a chemical reaction is same, whether it is brought about in two or more different ways in one or several steps. It is known as:

(A) Henry's law
(B) Joule's principle
(C) Hess's law
(D) Law of conservation of energy

Q14. For which system does the equilibrium constant, Kc has units of (concentration)^-1:

(A) N2(g) + 3H2(g) ⇌ 2NH3(g)
(B) H2(g) + I2(g) ⇌ 2HI(g)
(C) 2NO2(g) ⇌ N2O4(g)
(D) 2HF(g) ⇌ H2(g) + F2(g)

Q15. Molarity of pure water is:

(A) 1
(B) 18
(C) 55.5
(D) 6

Q16. If a strip of Cu metal is placed in a solution of FeSO4:

(A) Cu will be deposited
(B) Fe is precipitated out
(C) Cu and Fe both dissolve
(D) No reaction takes place

Q17. The unit of the rate constant is the same as that of the rate of reaction in:

(A) First order reaction
(B) Second order reaction
(C) Zero order reaction
(D) Third order reaction

## Essay Type (Group-II)

**Time Allowed:** 2.40 hrs. | **Maximum Marks:** 68

QSECTION – I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define Avogadro's number with a suitable example.

ii Write two assumptions of stoichiometry.

iii Many chemical reactions involve limiting reactant as taking place in our surrounding, justify.

iv Why crystallization is a better technique for separation and purification?

v Name any four sublimed solids.

vi What is Rf value; Also write its formula.

vii Deduce Boyle's law with the help of kinetic theory of gases.

viii Write any four applications of plasma.

ix The plot of PV versus P is a straight line at constant temperature and with a fix number of moles of an ideal gas, explain.

x How equilibrium constant (Kc) predicts direction of reaction.

xi How does buffer act?

xii Give optimum conditions to get maximum yield of Ammonia (NH3).

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Why gasoline evaporates faster than water?

ii Define molar heat of vapourization with example.

iii Define enthalpy change either it be positive or negative.

iv Differentiate between crystalline and amorphous solids. Give examples.

v What is frequency and wave number.

vi Define spectrum Give example.

vii What is atomic emission spectrum?

viii Differentiate between stark's effect and Xeemann's effect.

ix Define Roult's law with two statements giving its equation.

x Define solubility, give one example.

xi What is meant by promotor or activator, give example.

xii What is negative catalyst?

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Define electron affinity. Give an example.

ii The lone pair of electrons occupy more space than bond pair of electrons. Why?

iii Draw molecular orbital diagram of N2.

iv Write the basic assumption of Valence shell electron pair repulsion theory.

v What is a spontaneous process? Give an example.

vi Differentiate between internal energy and enthalpy.

vii Define enthalpy of formation. Give an example.

viii Impure copper can be purified by electrolytic process. How?

ix Differentiate between electrolytic cell and galvanic cell.

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QSECTION-II. (Marks 24) Note: Attempt any THREE questions.

QQ.5(a). Define Limiting reactant and how it is identified by different steps.

QQ.5(b). Explain Millikan's oil droplet experiment for determination of charge on electron.

QQ.6(a). There is a mixture of Hydrogen, helium and methane occupying a vessel of volume 13 dm^3 at 37°C and 1 atm. The masses of H2 and He are 0.8 g and 0.12 g respectively. Calculate the mole fraction of each gas?

QQ.6(b). What is Galvanic cell. Give the construction of Zn—Cu Galvanic cell. Briefly explain composition of electrodes and chemical reactions at electrodes.

QQ.7(a). Define electron Affinity. Enlist factors influencing Electron Affinity. How it varies in groups and periods?

QQ.7(b). State First law of thermodynamics. Prove that $\Delta E = qv$

QQ.8(a). What are liquid crystals? Give their four applications in daily life.

QQ.8(b). The solubility of CaF2 in water at 25°C is found to be 2.05 × 10^-4 mol dm^-3. Calculate the value of Ksp at this temperature.

QQ.9(a). Explain the depression of the freezing point of a solvent by a solute with the help of a graph.

QQ.9(b). What are enzymes? Give examples in which they act as catalyst. Mention the characteristics of enzyme catalysis.